Question	Answer	Marks	Partial Marks
1(a)	111	**3**	**M2** for $180 - \frac{180-42}{2}$ oe or $42 + \frac{180-42}{2}$ oe or **M1** for $\frac{180-42}{2}$ oe
1(b)	150	**3**	**M1** for $k \div (3 + 4 + 5)$ [$\times p$] where $p = 1, 3, 4$ or 5 or $\frac{5}{12}$ oe **B1** for 360 used
1(c)	$\frac{3}{5}$ cao nfww	**4**	**B3** for $\frac{72}{120}$ or **B2** for [d =] 72 or [h =] 120 or **M1** for 360 ÷ 5 oe isw or 180 – (360 ÷ 6) isw or for (6 – 2) × 180 [÷ 6]
1(d)	$x + 2x - 5 + x + 20 + 3x - 40 = 360$	**M1**	Accept equivalent equation e.g. $7x - 25 = 360$
	$7x = 360 + 5 - 20 + 40$ or better	**M1**	**FT** *their* equation, accept e.g. $7x = 385$
	$x = 55$	**B1**	
	55 and 125 or 105 and 75	**B1dep**	**Dep on M1M1B1** Accept 55 + 3 × 55 – 40 = 180 or 2 × 55 – 5 + 55 + 20 = 180 If B0 scored, **SC1** for 55, 75, 105 and 125
	Opposite angles sum to 180 oe [so *PQRS* is a cyclic quadrilateral]	**A1**	**Dep on M1M1B1B1**
1(e)	48.7 or 48.69 to 48.70…	**3**	**M2** for $\frac{360-50}{360} \times 2 \times \pi \times 9$ oe or **M1** for $\frac{50}{360} \times 2 \times \pi \times 9$ oe
2(a)	249.98 to 250[.0…]	**3**	**M2** for 830 – 500 × 1.16 or **M1** for 500 × 1.16 OR **M1** for 830 ÷ 1.16 **M1** for (*their* 715.5… – 500) × 1.16

Question	Answer	Marks	Partial Marks
2(b)(i)	33.5 or 33.51…	**2**	**M1** for $\frac{12\,400}{37\,000}$ [×100] oe If 0 scored, **SC1** for answer 66.5 or 66.48 to 66.49
2(b)(ii)	38 184 cao	**2**	**M1** for 37 000 × $\left(1+\frac{3.2}{100}\right)$ oe or **B1** for 1184
2(c)(i)	441 or 440.6 or 440.64 to 440.65	**3**	**B2** for answer 3941 or 3940.6 or 3940.64 to 3940.65 or **M2** for 3500 × $\left(1+\frac{2.4}{100}\right)^5$ – 3500 or **M1** for 3500 × $\left(1+\frac{2.4}{100}\right)^5$ oe isw
2(c)(ii)	16	**3**	**B2** for 15[.0] nfww to 15.1 or **M2** for 3500 × $\left(1+\frac{2.4}{100}\right)^{15}$ oe seen or 3500 × $\left(1+\frac{2.4}{100}\right)^{16}$ oe seen or **M1** for (3500 or *their* 3941) × $\left(1+\frac{2.4}{100}\right)^{n}$ associated with 5000 oe
3(a)(i)	$\frac{(x+3)(2x+5)}{2}=60$	**M1**	Accept $(x + 3)(2x + 5) = 2 \times 60$ or 120 Accept e.g. $(x + 3)(x + 2.5) = 60$ without division by 2 shown for M1 (but not A1)
	$2x^2 + 6x + 5x + 15$ seen	**B1**	Accept $2x^2 + 11x + 15$ seen
	$2x^2 + 11x – 105 = 0$	**A1**	Correct completion after M1B1 with the fraction seen removed with no errors or omissions seen
3(a)(ii)	$(2x + 21)(x – 5)$ [= 0]	**M2**	**M1** for partial factors $2x(x – 5) + 21(x – 5)$ [= 0] or $x(2x + 21) – 5(2x + 21)$ [= 0] OR $(2x + a)(x + b)$ [= 0] where $ab = – 105$ or $2b + a = 11$
	–10.5 and 5	**B1**	